

Question	Marking Guidance	Mark	Comments
10.1	1. Homogenise (tissue) to break open cells OR Homogenise (tissue) to release organelles/nuclei; 2. Filter to remove (intact) tissue/cells/debris; 3. Cold (solution) to prevent enzyme activity; 4. (Solution with) equivalent water potential to prevent osmosis OR (Solution with) equivalent water potential to prevent organelles bursting/shrinking; 5. Buffered (solution) to stop enzymes/protein denaturing; 6. Centrifuge/spin at low(er) speed so <u>nuclei</u> in pellet/move to bottom OR Centrifuge at low(er) speed and supernatant/solution discarded;	6 (6 x AO1)	1. Accept blend OR grind OR chop for homogenise 3. Accept 'slow down' for prevent 4. Accept isotonic for 'equivalent water potential' 4. Reject if reference is made to cells 6. Accept up to 1000 revolutions min⁻¹ OR 1000 × gravity for identified slow spin speed

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10.2	1. DNA in nucleus codes for enzyme/protein (production); 2. Ribosomes/rough endoplasmic reticulum produce enzyme/protein OR Translation on ribosomes/rough endoplasmic reticulum; 3. Rough endoplasmic reticulum transports/modifies/processes enzymes/protein; 4. Mitochondria produce ATP; 5. Golgi apparatus modify/process/package/transport enzymes/protein OR Golgi apparatus make/transport glycoprotein OR Golgi apparatus forms/releases vesicles; 6. Vesicles move (protein) to cell(-surface) membrane OR Vesicles fuse with cell(-surface) membrane;	5 max (5 x AO1)	1. Accept genetic material/code OR gene for DNA Accept polypeptide for protein/enzyme 3. Ignore rER/RER only once 4. Reject produce energy 5. Accept body for apparatus 5 Accept 'adds lipid/carbohydrate to' for modify 5. Accept lipoprotein for glycoprotein 5 and 6 Accept lysosome for vesicle 6. Accept exocytosis for 'fuse with cell membrane'

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10.3	1. Ribose, Adenine and 3 phosphates; 2. ATP to ADP + Pi by ATP hydrolase in hydrolysis (reaction); 3. ADP + Pi to ATP by ATP synthase; 4. (In) condensation (reaction);	4 (4 x AO1)	1. Accept a labelled diagram showing ribose, adenine and 3 phosphates 1. Accept adenosine and 3 phosphates 1. Reject Adenosine and 3 phosphates if ribose/pentose is also mentioned 1. Ignore pentose sugar 2. Accept ATPase for ATP hydrolase 2. Accept hydrolayse 3. Accept synthayse